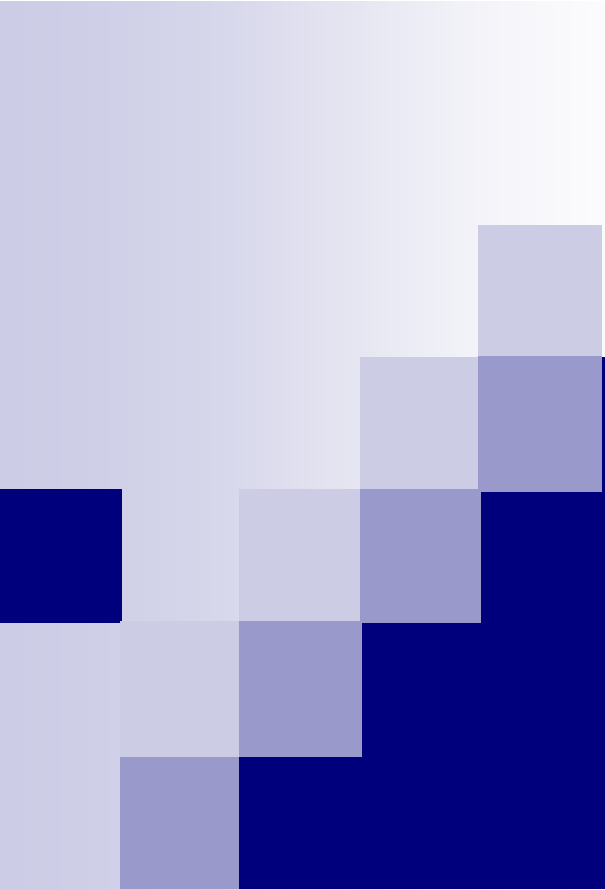
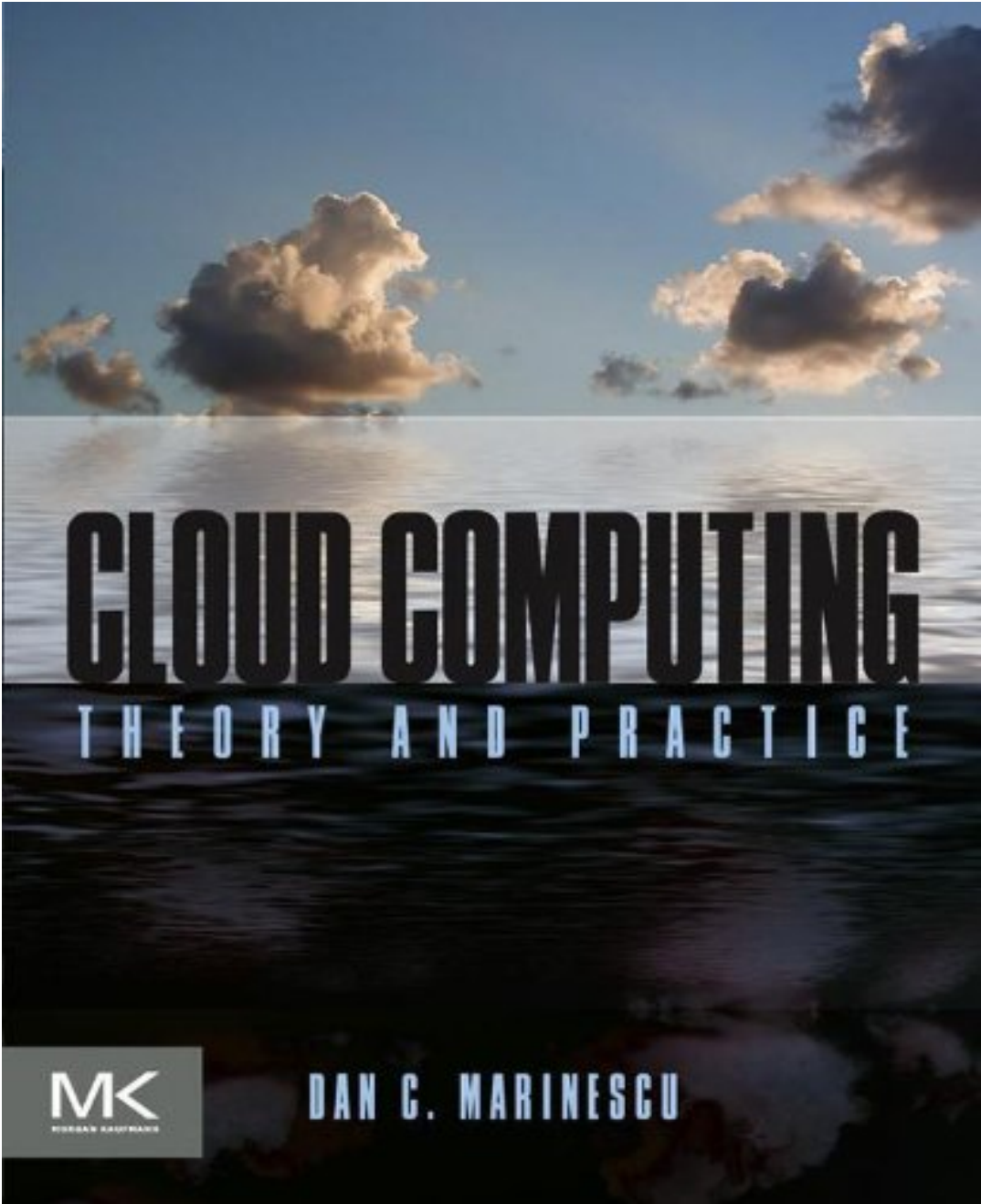




# Cloud Infrastructure



# CLOUD COMPUTING

THEORY AND PRACTICE

**MK**  
MORGAN KAUFMANN

DAN C. MARINESCU



# Contents

- IaaS services from Amazon.
  - Regions and availability zones for Amazon Web Services.
  - Instances – attributes and cost.
  - A repertoire of Amazon Web Services.
- SaaS and PaaS services from Google.
- SaaS and PaaS services from Microsoft.
- Open-source platforms for private clouds.
- Cloud storage diversity and vendor lock-in.
- Cloud interoperability; the Intercloud.
- Energy use and ecological impact large datacenters .
- Service and compliance level agreements.
- Responsibility sharing between user and the cloud service provider.
- User security concerns.
- User motivation.



# Who is Amazon !!

American international multibillion dollar electronic commerce company with headquarters in Seattle, Washington, USA.

- started in 1995 by Jeff Bezos as an online bookstore.
- but soon diversified, selling DVDs, VHSs, CDs, video and MP3 downloads/streaming, software, video games, electronics, apparel, furniture, food, toys, and jewelry.
- The company also produces consumer electronics: Kindle e-book reader and the Kindle Fire tablet computer.
- In 2006, Amazon officially launched the Amazon Web Services (AWS) to become a major provider of cloud computing services.



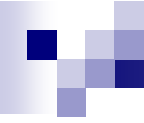
# Amazon Web Services (AWS)

- AWS IaaS cloud computing services launched in 2006.
- Businesses in 200 countries used AWS in 2012.
- The infrastructure consists of compute and storage servers interconnected by high-speed networks and supports a set of services.
- An application developer:
  - Installs applications on a platform of his/her choice.
  - Manages resources allocated by Amazon.



# Amazon Web Services (AWS)

- The introduction by Amazon of its Elastic Compute Cloud (EC2) service in 2006 marked the true beginning of cloud computing.
- EC2 is based on virtualization technology, which allows one server to run independent, isolated operating systems (OSs) for multiple users simultaneously.
- Since then Microsoft, Google, and many others have introduced virtual machine (VM) services based on this technology



# Some Major cloud infrastructure

- The cloud computing infrastructure at Amazon, Google, and Microsoft (as of mid 2012).
  - Amazon is a pioneer in Infrastructure-as-a-Service (IaaS).
  - Google's efforts are focused on Software-as-a-Service (SaaS) and Platform-as-a-Service (PaaS).
  - Microsoft is involved in PaaS.
- Private clouds are an alternative to public clouds. Open-source cloud computing platforms such as:
  - Eucalyptus,
  - OpenNebula,
  - Nimbus,
  - OpenStack

can be used as a control infrastructure for a private cloud.

# AWS regions and availability zones

- Amazon offers cloud services through a network of data centers on several continents.
- In each *region* there are several *availability zones* interconnected by high-speed networks.
- An *availability zone* is a data center consisting of a large number of servers.

| Region        | Location          | Availability zones  | Cost      |
|---------------|-------------------|---------------------|-----------|
| US West       | Oregon            | us-west-2a/2b/2c    | Low       |
| US West       | North California  | us-west-1a/1b/1c    | High      |
| US East       | North Virginia    | us-east-1a/2a/3a/4a | Low       |
| Europe        | Ireland           | eu-west-1a/1b/1c    | Medium    |
| South America | Sao Paulo, Brazil | sa-east-1a/1b       | Very high |
| Asia Pacific  | Tokyo, Japan      | ap-northeast-1a/1b  | High      |
| Asia Pacific  | Singapore         | ap-southeast-1a/1b  | Medium    |

- Regions do not share resources and communicate through the Internet.





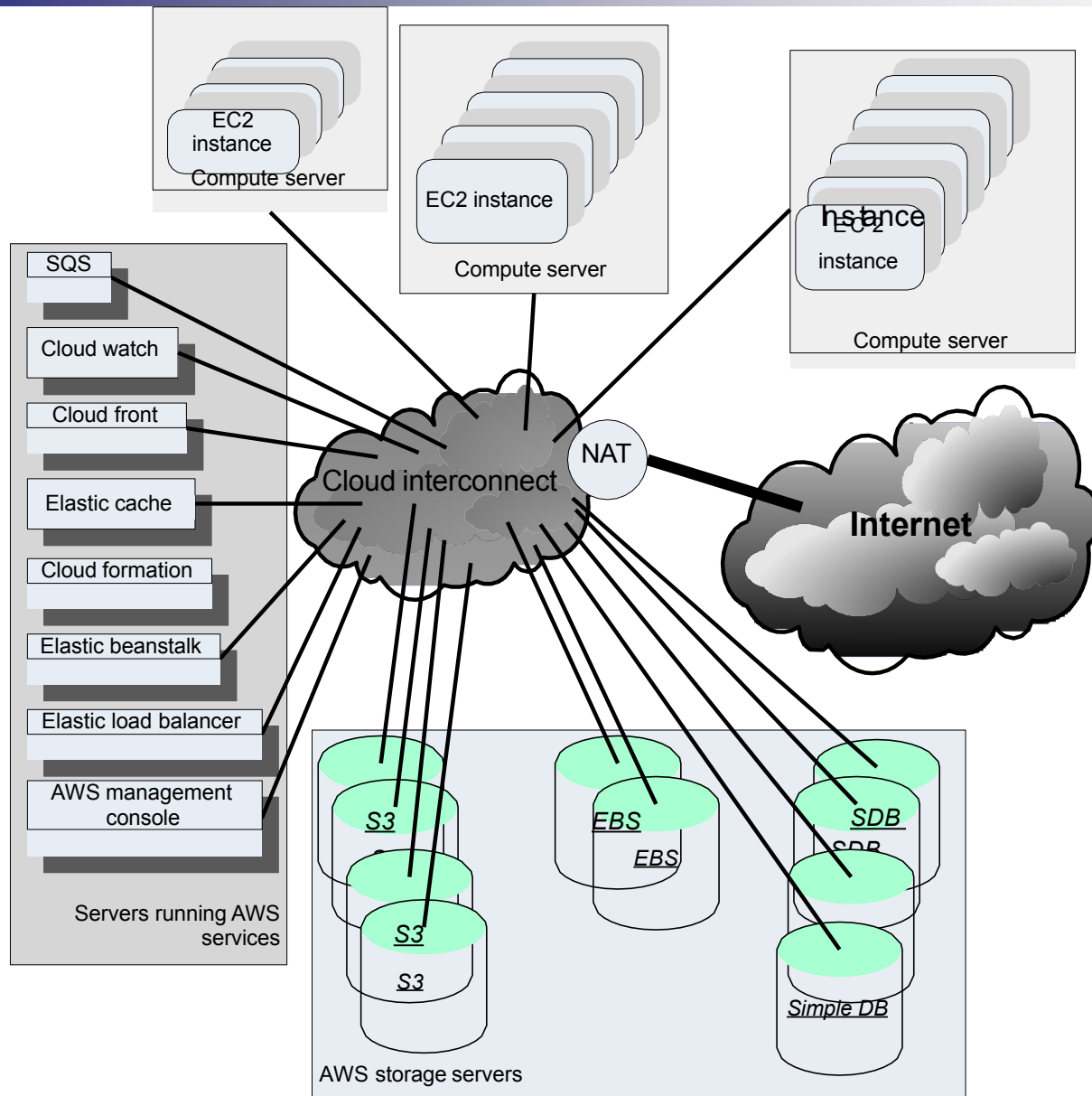
# AWS instances

- An instance is a virtual server with a well specified set of resources including: CPU cycles, main memory, secondary storage, communication and I/O bandwidth.
- The user chooses:
  - The region and the availability zone where this virtual server should be placed.
  - An instance type from a limited menu of instance types.
- When launched, an instance is provided with a DNS name; this name maps to a
  - *private IP address*: for internal communication within the internal EC2 communication network.
  - *public IP address*: for communication outside the internal Amazon network, e.g., for communication with the user that launched the instance.

CPU cycle Usually, the time required for the execution of one simple processor operation such as an addition; this time is normally the reciprocal of the clock rate.

# AWS instances (cont'd)

- Network Address Translation (NAT) maps external IP addresses to internal ones.
- The public IP address is assigned for the lifetime of an instance.
- An instance can request an *elastic IP address*, rather than a public IP address. The elastic IP address is a static public IP address allocated to an instance from the available pool of the availability zone.
- An elastic IP address is not released when the instance is stopped or terminated and must be released when no longer needed. (Else ?)





# Steps to run an application

- Retrieve the user input from the front-end.
- Retrieve the disk image of a VM (Virtual Machine) from a repository.
- Locate a system and requests the VMM (Virtual Machine Monitor) running on that system to setup a VM.
- Invoke the Dynamic Host Configuration Protocol (DHCP) and the IP bridging software to set up MAC and IP addresses for the VM.



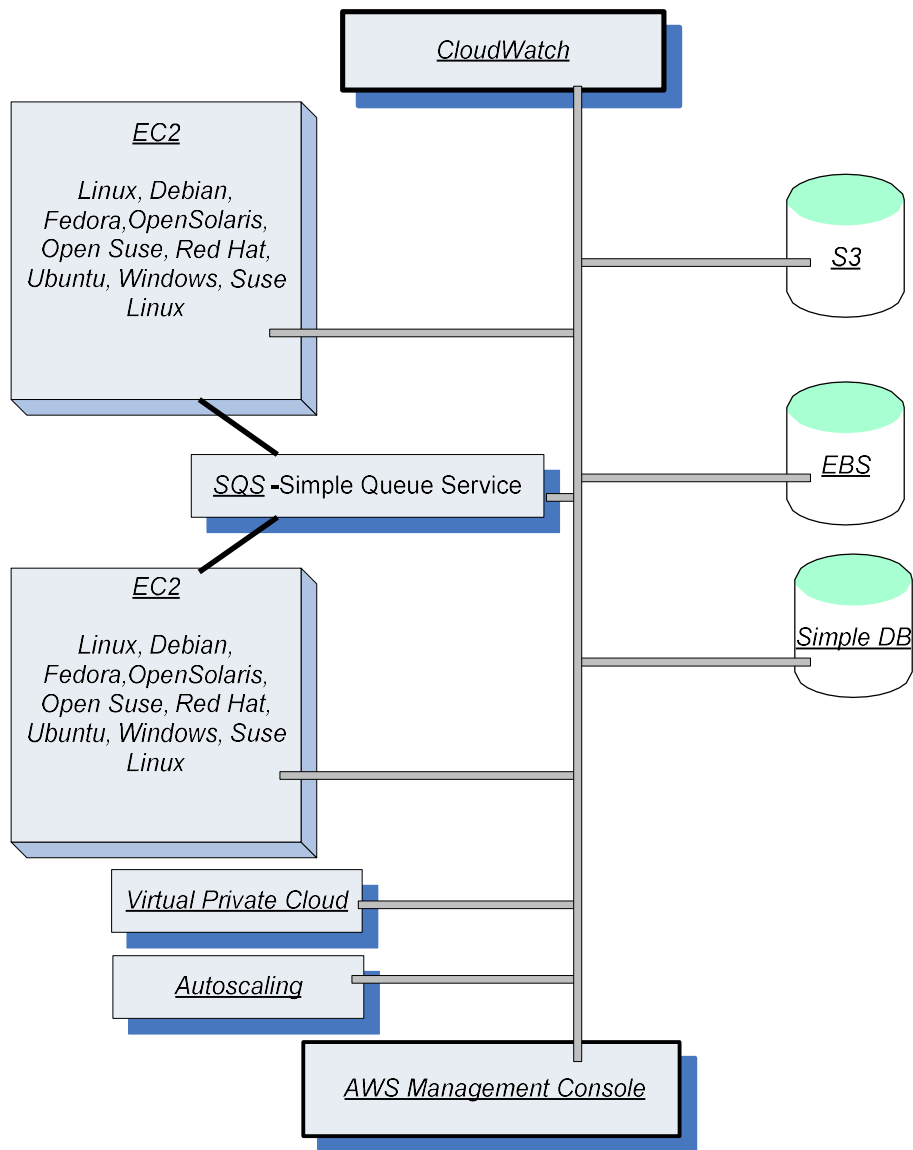
# User interactions with AWS

- The AWS Management Console. The easiest way to access all services, but not all options may be available.
- AWS SDK libraries and toolkits are provided for several programming languages including Java, PHP, C#, and Objective-C.



# Examples of Amazon Web Services

- *AWS Management Console* - allows users to access the services offered by AWS .
- *Elastic Cloud Computing (EC2)* - allows a user to launch a variety of operating systems.
- *Simple Queuing Service (SQS)* - allows multiple *EC2* instances to communicate with one another.
- *Simple Storage Service (S3), Simple DB, and Elastic Bloc Storage (EBS)* - storage services.
- *Cloud Watch* - supports performance monitoring.
- *Auto Scaling* - supports elastic resource management.
- *Virtual Private Cloud* - allows direct migration of parallel applications.
- And many more....





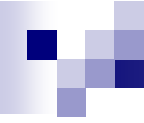


# EC2 – Elastic Cloud Computing

- *EC2* - web service for launching instances of an application under several operating systems, such as:
  - Several Linux distributions.
  - Microsoft Windows Server 2003 and 2008.
  - OpenSolaris.
  - FreeBSD.
  - NetBSD.
- A user can
  - Load an *EC2* instance with a custom application environment.
  - Manage network's access permissions.
  - Run the image using as many or as few systems as desired.

## EC2 (cont'd)

- Import virtual machine (VM) images from the user environment to an instance through *VM import*.
- *EC2* instances boot from an AMI (Amazon Machine Image) digitally signed and stored in S3.
- Users can access:
  - Images provided by Amazon.
  - Customize an image and store it in S3.
- An *EC2* instance is characterized by the resources it provides:
  - VC (Virtual Computers) – virtual systems running the instance.
  - CU (Compute Units) – measure computing power of each system.
  - Memory.
  - I/O capabilities.



# Instance types

- Standard instances: micro (StdM), small (StdS), large (StdL), extra large (StdXL); small is the default.
- High memory instances: high-memory extra large (HmXL), high-memory double extra large (Hm2XL), and high-memory quadruple extra large (Hm4XL).
- High CPU instances: high-CPU extra large (HcpuXL).
- Cluster computing: cluster computing quadruple extra large (Cl4XL).

# Instance types

| Instance name | API name    | Platform (32/64-bit) | Memory (GB) | Max EC2 compute units | I-memory (GB) | I/O (M/H) |
|---------------|-------------|----------------------|-------------|-----------------------|---------------|-----------|
| StdM          |             | 32 and 64            | 0. 633      | 1 VC; 2 CUs           |               |           |
| StdS          | m1.small    | 32                   | 1.7         | 1 VC; 1 CU            | 160           | M         |
| StdL          | m1.large    | 64                   | 7.5         | 2 VCs; 2 × 2 CUs      | 85            | H         |
| StdXL         | m1.xlarge   | 64                   | 15          | 4 VCs; 4 × 2 CUs      | 1,690         | H         |
| HmXL          | m2.xlarge   | 64                   | 17.1        | 2 VCs; 2 × 3.25 CUs   | 420           | M         |
| Hm2XL         | m2.2xlarge  | 64                   | 34.2        | 4 VCs; 4 × 3.25 CUs   | 850           | H         |
| Hm4XL         | m2.4xlarge  | 64                   | 68.4        | 8 VCs; 8 × 3.25 CUs   | 1,690         | H         |
| HcpuXL        | c1.xlarge   | 64                   | 7           | 8 VCs; 8 × 2.5 CUs    | 1,690         | H         |
| Cl4XL         | cc1.4xlarge | 64                   | 18          | 33.5 CUs              | 1,690         | H         |

# Instance cost

- A main attraction of the Amazon cloud computing is the low cost.

| Instance | Linux/Unix | Windows |
|----------|------------|---------|
| StdM     | 0.007      | 0.013   |
| StdS     | 0.03       | 0.048   |
| StdL     | 0.124      | 0.208   |
| StdXL    | 0.249      | 0.381   |
| HmXL     | 0.175      | 0.231   |
| Hm2XL    | 0.4        | 0.575   |
| Hm4XL    | 0.799      | 1.1     |
| HcpuXL   | 0.246      | 0.516   |
| C14XL    | 0.544      | N/A     |



# Amazon EC2 Functionality

- Select a pre-configured, templated image to get up and running immediately.
- Or create an Amazon Machine Image (AMI) containing your applications, libraries, data, and associated configuration settings.
- Configure security and network access on your Amazon EC2 instance.
- Choose which instance type(s) and operating system you want, then start, terminate, and monitor as many instances of your AMI as needed, using the web service APIs or the variety of management tools provided.
- Determine whether you want to run in multiple locations, utilize static IP endpoints, or attach persistent block storage to your instances.
- Pay only for the resources that you actually consume, like instance-hours or data transfer.

# OS and Software

**Amazon Machine Images (AMIs)** are preconfigured with an ever-growing list of operating systems.

| Operating Systems                        |                                  |                                         |
|------------------------------------------|----------------------------------|-----------------------------------------|
| <a href="#">Red Hat Enterprise Linux</a> | <a href="#">Windows Server</a>   | <a href="#">Oracle Enterprise Linux</a> |
| <a href="#">SUSE Linux Enterprise</a>    | <a href="#">Amazon Linux AMI</a> | Ubuntu Linux                            |
| Fedora                                   | Gentoo Linux                     | Debian                                  |

| Databases                                     | Resource Management            | Web Hosting                                      |
|-----------------------------------------------|--------------------------------|--------------------------------------------------|
| <a href="#">IBM DB2</a>                       | <a href="#">StackIQ Rocks+</a> | Apache HTTP                                      |
| <a href="#">IBM Informix Dynamic Server</a>   | Hadoop                         | IIS/Asp.Net                                      |
| <a href="#">Microsoft SQL Server Standard</a> | Condor                         | <a href="#">IBM Lotus Web Content Management</a> |
| <a href="#">MySQL Enterprise</a>              |                                | <a href="#">IBM WebSphere Portal Server</a>      |
| <a href="#">Oracle Database 11g</a>           |                                |                                                  |

| Application Development Environments                  | Application Servers                              | Video Encoding & Streaming             |
|-------------------------------------------------------|--------------------------------------------------|----------------------------------------|
| <a href="#">IBM sMash</a>                             | <a href="#">IBM WebSphere Application Server</a> | <a href="#">Wowza Media Server Pro</a> |
| <a href="#">JBoss Enterprise Application Platform</a> | Java Application Server                          | Windows Media Server                   |
| Ruby on Rails                                         | <a href="#">Oracle WebLogic Server</a>           |                                        |



# Security

- Amazon EC2 includes web service interfaces to configure firewall settings that control network access to and between groups of instances.
- When launching Amazon EC2 resources within [Amazon Virtual Private Cloud](#) (Amazon VPC), you can isolate your compute instances by specifying the IP range you wish to use, and connect to your existing IT infrastructure using industry-standard encrypted IPsec VPN.
- You can also choose to launch [Dedicated Instances](#) into your VPC. Dedicated Instances are Amazon EC2 Instances that run on hardware dedicated to a single customer for additional isolation.





# Cost Saving Considerations

**On-Demand Instances** – On-Demand Instances let you pay for compute capacity by the hour with no long-term commitments. This frees you from the costs and complexities of planning, purchasing, and maintaining hardware and transforms what are commonly large fixed costs into much smaller variable costs.

**Reserved Instances** – Reserved Instances give you the option to make a low, one-time payment for each instance you want to reserve and in turn receive a significant discount on the hourly charge for that instance.

**Spot Instances** – Spot Instances allow customers to bid on unused Amazon EC2 capacity and run those instances for as long as their bid exceeds the current Spot Price.

# S3 – Simple Storage System

- Service designed to store large objects; an application can handle an unlimited number of objects ranging in size from 1 byte to 5 TB.
- An object is stored in a bucket and retrieved via a unique, developer-assigned key; a bucket can be stored in a Region selected by the user.
- Supports a minimal set of functions: write, read, and delete; it does not support primitives to copy, to rename, or to move an object from one bucket to another.
- The object names are global.
- S3 maintains for each object: the name, modification time, an access control list, and up to 4 KB of user-defined metadata.

## S3 (cont'd)

- Authentication mechanisms ensure that data is kept secure.
- Objects can be made public, and rights can be granted to other users.
- S3 computes the MD5 of every object written and returns it in a field called ETag.
- System computes the MD5 of an object stored or written to compare this with the ETag; if the two values do not match, then the object was corrupted during transmission or storage.

# Elastic Block Store (EBS)

- Provides persistent block level storage volumes for use with *EC2* instances; suitable for database applications, file systems, and applications using raw data devices.
- A volume appears to an application as a raw, unformatted and reliable physical disk; the range 1 GB -1 TB.
- An *EC2* instance may mount multiple volumes, but a volume cannot be shared among multiple instances.
- The volumes are grouped together in Availability Zones and are automatically replicated in each zone.
- EBS volumes are network-attached like SAN, and persist independently from the life of an instance.

Persistence: the fact of continuing in an opinion or course of action in spite of difficulty or opposition



# Elastic Block Store (EBS)

EBS provides the ability to **create point-in-time snapshots of volumes**, which are **persisted to Amazon S3**.

- These snapshots can be **used as the starting point for new Amazon EBS volumes**, and protect data for long-term durability.
- The **same snapshot can be used to instantiate as many volumes as you wish**.
- These snapshots **can be copied across AWS regions**.



# EBS and S3

- EBS can only be used with EC2 instances while S3 can be used outside EC2
- EBS appears as a mountable volume while the S3 requires software to read and write data
- EBS can accommodate a smaller amount of data than S3
- EBS can only be used by one EC2 instance at a time while S3 can be used by multiple instances
- S3 typically experiences write delays while EBS does not

# Amazon Elastic Beanstalk



<https://www.goodreads.com/book/show/18455358-jack-and-the-beanstalk>



# Amazon Elastic Beanstalk

**AWS Elastic Beanstalk** provides a solution to **quickly deploy** and **manage** applications in the AWS cloud.

Deploy using **AWS Management Console, a Git repository, or an integrated development environment (IDE) such as Eclipse or Visual Studio** to upload your application

You **simply upload your application**, and Elastic Beanstalk **automatically** handles the deployment details of capacity **provisioning, load balancing, auto-scaling**, and application health **monitoring**.

Elastic Beanstalk leverages AWS services such as **Amazon EC2, Amazon S3**, ....

To ensure easy portability of your application, Elastic Beanstalk is built using familiar software stacks such as:

- Apache HTTP Server for Node.js, PHP and Python
- Passenger for Ruby,
- IIS 7.5 for .NET
- Apache Tomcat for Java.

There is **no additional charge** for Elastic Beanstalk - you **pay only for the AWS resources** needed to store and run your applications.





# SimpleDB

- Non-relational data store. Supports store and query functions traditionally provided only by relational databases.
- Supports high performance Web applications; users can store and query data items via Web services requests.
- Creates multiple geographically distributed copies of each data item.
- It manages automatically:
  - The infrastructure provisioning.
  - Hardware and software maintenance.
  - Replication and indexing of data items.
  - Performance tuning.

- 
- SimpleDB would be used over S3 in the case where you want to store multiple key-value pairs associated with an item and want to retain the ability to find items based on any of the key-value pairs.
  - SimpleDB is more flexible with the stored metadata. The key-value pairs are indexed, so querying can be quick.
  - You can add and modify key-value pairs that are already in SimpleDB, where you'd need to delete and recreate objects in S3 to update metadata.



# Storage

- **Instance –store** : disappears with the instance (transient)
- **Block storage**: SAN-like, persists across time
- **S3** is independent of an instance: for archival purposes:  
vault: store it now and retrieve it at a later date
- **SimpleDB**: Relational database better than MySQL or Oracle for reliability.



# SQS - Simple Queue Service

- Amazon Simple Queue Service (SQS) is a fully managed message queuing service
- SQS eliminates the complexity and overhead associated with managing and operating message oriented middleware.
- Using SQS, user can send, store, and receive messages between software components at any volume, without losing messages or requiring other services to be available.
- User can get started with SQS in minutes using the AWS console, Command Line Interface or SDK of your choice, and three simple commands.



# SQS - Simple Queue Service

- Hosted message queues are accessed through standard Simple Object Access Protocol (SOAP) and Query interfaces.
- Supports automated workflows - *EC2* instances can coordinate by sending and receiving SQS messages.
- Applications using SQS can run independently and asynchronously, and do not need to be developed with the same technologies.
- User can use Amazon SQS to exchange sensitive data between applications using server-side encryption (SSE) to encrypt each message body.
- A received message is “locked” during processing; if processing fails, the lock expires and the message is available again.
- Queue sharing can be restricted by IP address and time-of-day.



# SQS - Simple Queue Service

- SQS offers two types of message queues.
  - Standard queues offer maximum throughput, best-effort ordering, and at-least-once delivery.
  - SQS FIFO queues are designed to guarantee that messages are processed exactly once, in the exact order that they are sent.

# Amazon Simple Workflow Service (SWF)

Amazon SWF is a **task coordination** and **state management service** for cloud applications.

Using Amazon SWF, user **structures** the various **processing steps** in an application that **runs across one or more machines** as a set of “**tasks**.”

Amazon SWF **manages dependencies** between the tasks, **schedules** the tasks for execution, and runs any logic that needs to be **executed in parallel**.

The service also **tracks** the tasks’ **progress**.

As the **business requirements change**, Amazon SWF makes it **easy to change application logic** without having to **worry about the underlying state machinery and flow control**.



# Amazon Virtual Private Cloud (VPC)

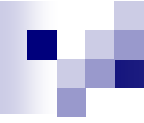
**Amazon VPC** lets you **provision** a **logically isolated section** of the Amazon Web Services (AWS) Cloud.

You have **complete control** over your virtual networking environment, including:  
selection of your own **IP address range**,  
**creation of subnets**, and  
**configuration of route tables** and **network gateways**.

VPC allows **bridging with an onsite IT infrastructure** with an **encrypted VPN connection** with an **extra charge per VPN Connection-hour**.

There is **no additional charge** for using Amazon Virtual Private Cloud, aside from the normal Amazon EC2 usage charges.





# CloudWatch

- Monitoring infrastructure used by application developers, users, and system administrators to collect and track metrics important for optimizing the performance of applications and for increasing the efficiency of resource utilization.
- Without installing any software a user can monitor either seven or eight pre-selected metrics and then view graphs and statistics for these metrics.
- When launching an Amazon Machine Image (AMI) the user can start the CloudWatch and specify the type of monitoring:
  - Basic Monitoring - free of charge; collects data at five-minute intervals for up to seven metrics.
  - Detailed Monitoring - subject to charge; collects data at one minute interval.

# AWS services introduced in 2012

- *Route 53* - low-latency DNS service used to manage user's DNS public records.
- *Elastic MapReduce (EMR)* - supports processing of large amounts of data using a hosted Hadoop running on *EC2*.
- *Simple Workflow Service (SWF)* - supports workflow management; allows scheduling, management of dependencies, and coordination of multiple *EC2* instances.
- *ElastiCache* - enables web applications to retrieve data from a managed in-memory caching system rather than a much slower disk-based database.
- *DynamoDB* - scalable and low-latency fully managed NoSQL database service.
- out More.....Assignment: Make ppt's about AWS latest

# AWS services introduced in 2012 (cont'd)

- *CloudFront* - web service for content delivery.
- *Elastic Load Balancer* - automatically distributes the incoming requests across multiple instances of the application.
- *Elastic Beanstalk* - handles automatically deployment, capacity provisioning, load balancing, auto-scaling, and application monitoring functions.
- *CloudFormation* - allows the creation of a stack describing the infrastructure for an application.
- Find out more (Assignment 1)





# SaaS services offered by Google

- *Gmail* - hosts Emails on Google servers and provides a web interface to access the Email.
- *Google docs* - a web-based software for building text documents, spreadsheets and presentations.
- *Google Calendar* - a browser-based scheduler; supports multiple user calendars, calendar sharing, event search, display of daily/weekly/monthly views, and so on.
- *Google Groups* - allows users to host discussion forums to create messages online or via Email.
- *Picasa* - a tool to upload, share, and edit images.
- *Google Maps* - web mapping service; offers street maps, a route planner, and an urban business locator for numerous countries around the world



# PaaS services offered by Google

- *AppEngine* - a developer platform hosted on the cloud.
  - Initially supported Python, Java was added later.
  - The database for code development can be accessed with GQL (Google Query Language) with a SQL-like syntax.
- *Google Co-op* - allows users to create customized search engines based on a set of facets/categories.
- *Google Drive* - an online service for data storage.
- *Google Base* - allows users to load structured data from different sources to a central repository, a very large, self-describing, semi-structured, heterogeneous database.



# PaaS and SaaS services from Microsoft

- *Windows Azure* - an operating system; has 3 components:
  - Compute - provides a computation environment.
  - Storage - for scalable storage.
  - Fabric Controller - deploys, manages, and monitors applications.
- *SQL Azure* - a cloud-based version of the SQL Server.
- *Azure AppFabric*, formerly .NET Services - a collection of services for cloud applications.

# Open-source platforms for private clouds

- *Eucalyptus* - can be regarded as an open-source counterpart of Amazon's EC2.
- *Open-Nebula* - a private cloud with users actually logging into the head node to access cloud functions. The system is centralized and its default configuration uses the NFS file system.
- *Nimbus* - a cloud solution for scientific applications based on Globus software; inherits from Globus:
  - The image storage.
  - The credentials for user authentication.
  - The requirement that a running Nimbus process can **ssh** into all compute nodes.
- Assignment 2: Write up on Eucalyptus, Openstack, Nimbus



# Eucalyptus

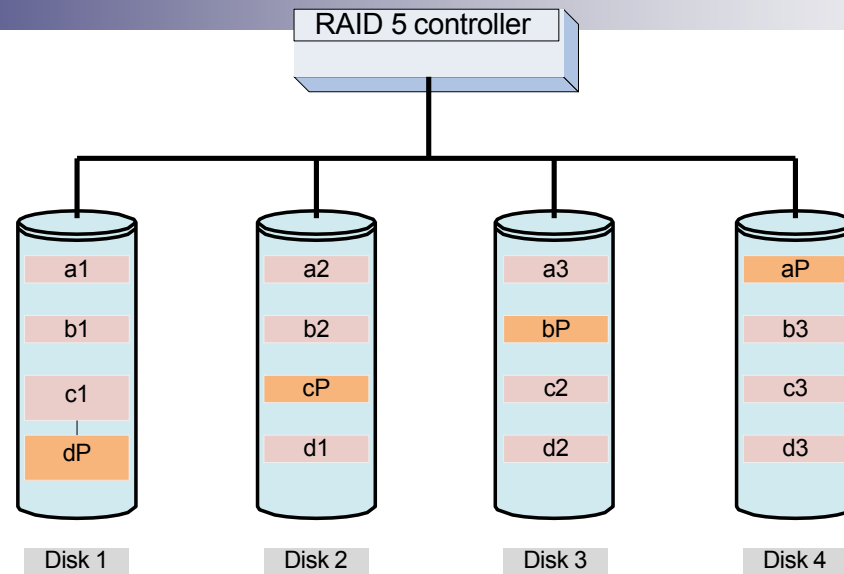
- *Virtual Machines* - run under several VMMs including Xen, KVM, and VMware.
- *Node Controller* - runs on server nodes hosting a VM and controls the activities of the node.
- *Cluster Controller* - controls a number of servers.
- *Cloud Controller* - provides the cloud access to end-users, developers, and administrators.
- *Storage Controller* - provides persistent virtual hard drives to applications. It is the correspondent of EBS.
- *Storage Service (Walrus)* - provides persistent storage; similar to S3, it allows users to store objects in buckets.



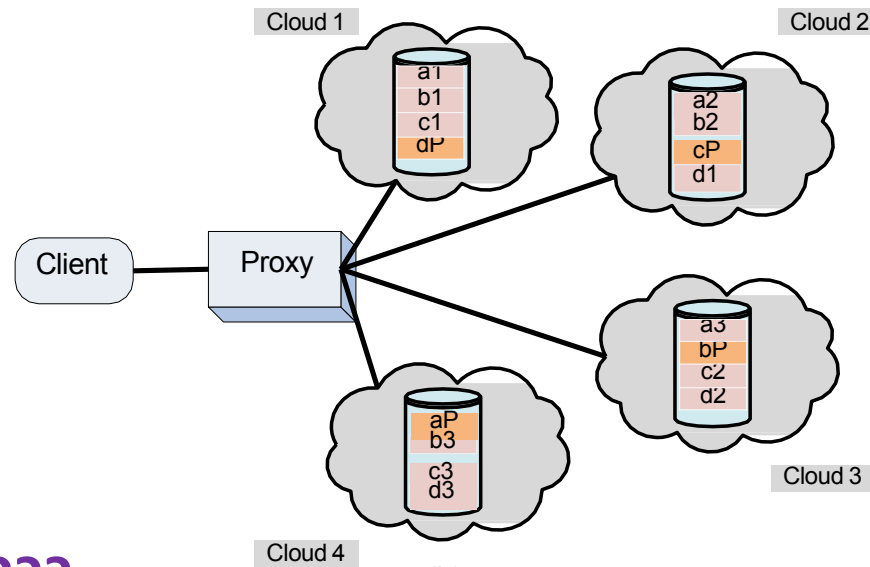


# Cloud storage diversity and vendor lock-in

- Risks when a large organization relies on a single cloud service provider:
  - Cloud services may be unavailable for a short or an extended period of time.
  - Permanent data loss in case of a catastrophic system failure.
  - The provider may increase the prices for service.
- Switching to another provider could be very costly due to the large volume of data to be transferred from the old to the new provider.
- A solution is to replicate the data to multiple cloud service providers, similar to data replication in RAID



(a)



(b)

.....Comments ???



# Cloud interoperability; the Intercloud

- An Intercloud is a federation of clouds that cooperate to provide a better user experience.
- Is an Intercloud feasible?
- Not likely at this time:
  - There are no standards for either storage or processing.
  - The clouds are based on different delivery models.
  - The set of services supported by these delivery models is large and open; new services are offered every few months.
  - CSPs (Cloud Service Providers) believe that they have a competitive advantage due to the uniqueness of the added value of their services.
  - Security is a major concern for cloud users and an Intercloud could only create new threats.

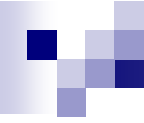
# Energy use and ecological impact

- The energy consumption of large-scale data centers and their costs for energy and for cooling are significant.
- In 2006, the 6,000 data centers in the U.S consumed  $61 \times 10^9$  KWh of energy, 1.5% of all electricity consumption, at a cost of \$4.5 billion.
- The energy consumed by the data centers was expected to double from 2006 to 2011 and peak instantaneous demand to increase from 7 GW to 12 GW.
- The greenhouse gas emission due to the data centers is estimated to increase from  $116 \times 10^9$  tones of  $\text{CO}_2$  in 2007 to 257 tones in 2020 due to increased consumer demand.
- The effort to reduce energy use is focused on computing, networking, and storage activities of a data center.



## Energy use and ecological impact (cont'd)

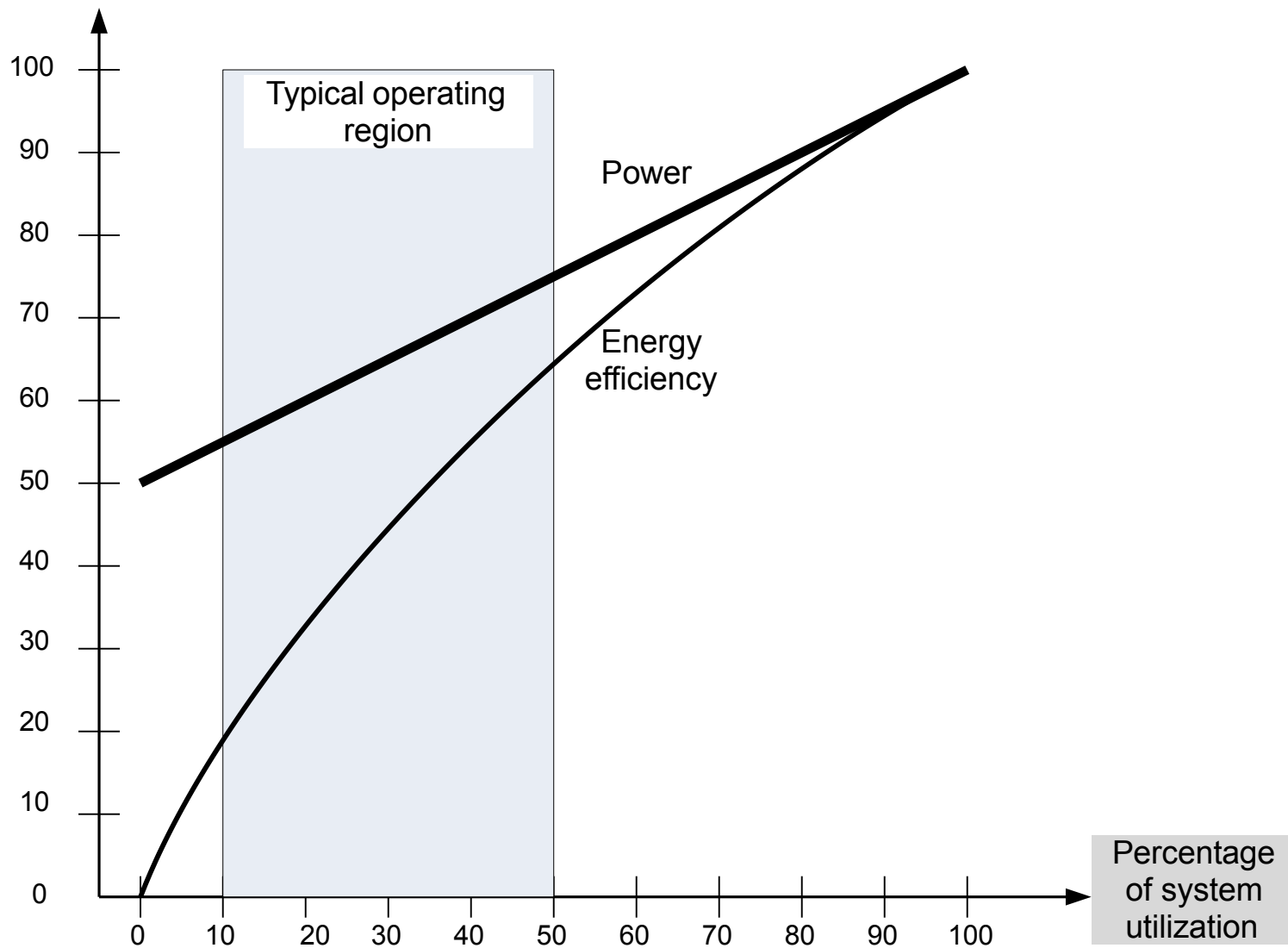
- Operating efficiency of a system is captured by the *performance per Watt of power*.
- The performance of supercomputers has increased 3.5 times faster than their operating efficiency – 7,000% versus 2,000% during the period 1998 – 2007.
- A typical Google cluster spends most of its time within the 10-50% CPU utilization range; there is a mismatch between server workload profile and server energy efficiency.



# Energy-proportional systems

- An energy-proportional system consumes no power when idle, very little power under a light load and, gradually, more power as the load increases.
- By definition, an ideal energy-proportional system is always operating at 100% efficiency.
- Humans are a good approximation of an ideal energy proportional system; about 70 W at rest, 120 W on average on a daily basis, and can go as high as 1,000 – 2,000 W during a strenuous, short time effort.
- Even when power requirements scale linearly with the load, the energy efficiency of a computing system is not a linear function of the load; even when idle, a system may use 50% of the power corresponding to the full load.

Percentage of  
power usage



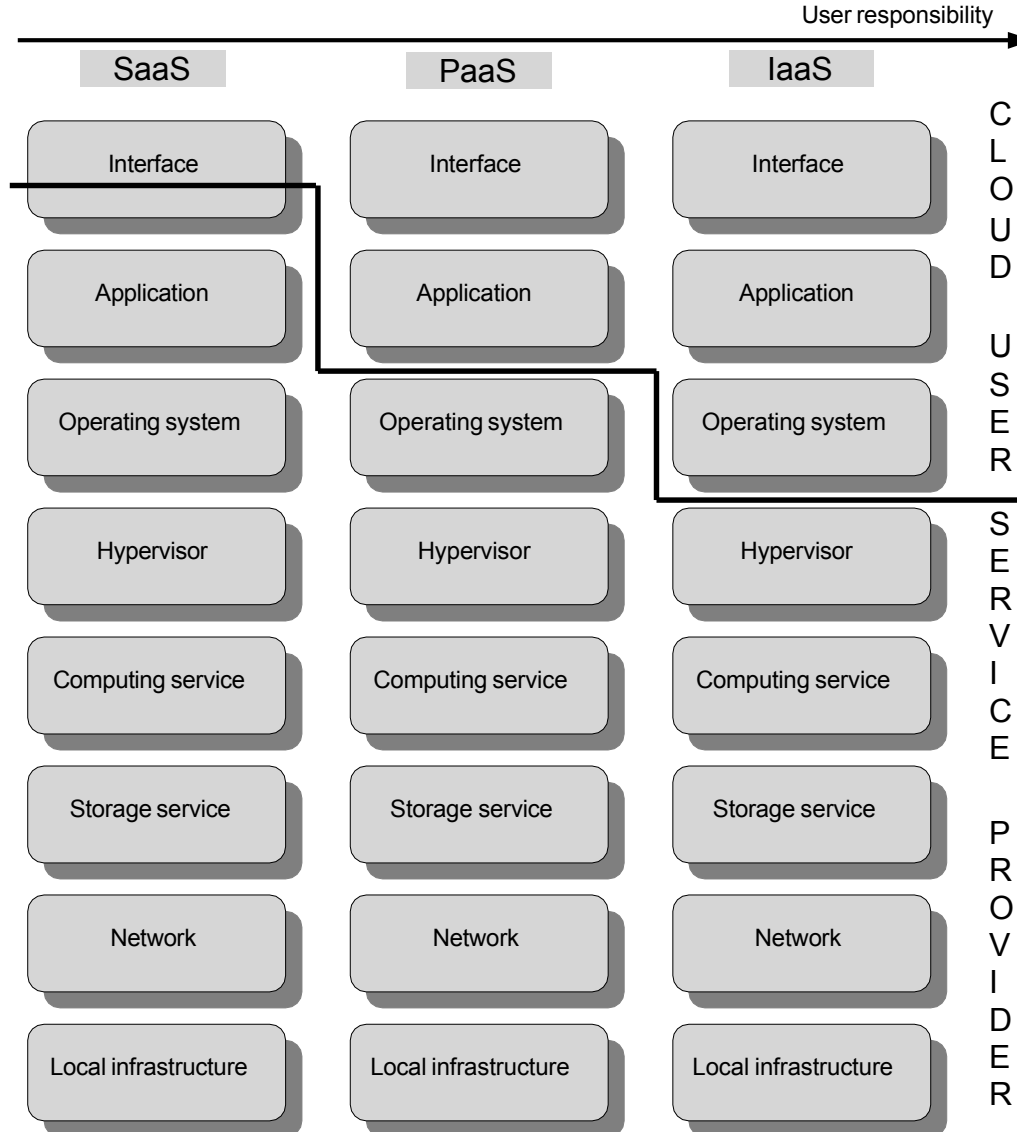


# Service Level Agreement (SLA)

- SLA - a negotiated contract between the customer and CSP; can be legally binding or informal. Objectives:
  - Identify and define the customer's needs and constraints including the level of resources, security, timing, and QoS.
  - Provide a framework for understanding; a critical aspect of this framework is a clear definition of classes of service and the costs.
  - Simplify complex issues; clarify the boundaries between the responsibilities of clients and CSP in case of failures.
  - Reduce areas of conflict.
  - Encourage dialog in the event of disputes.
  - Eliminate unrealistic expectations.
- Specifies the services that the customer receives, rather than how the cloud service provider delivers the services.



# Responsibility sharing between user and CSP





# User security concerns

- Potential loss of control/ownership of data.
- Data integration, privacy enforcement, data encryption.
- Data remanence after de-provisioning.
- Multi tenant data isolation.
- Data location requirements within national borders.
- Hypervisor security.
- Audit data integrity protection.
- Verification of subscriber policies through provider controls.
- Certification/Accreditation requirements for a given cloud service.

# Preliminaries

We will use Microsoft's Azure for Hands-on Exercises

Initializing your account

Download Azure Storage Explorer <http://storageexplorer.com/>

Access tutorial tarball <https://SciEngCloud.github.io/tutorial.tar.gz>

On mac:

```
gunzip tutorial.tar.gz
```

```
tar -xf tutorial.tar
```

on PC use something like 7-zip or other decompression and extractor.

Access to these slides and Jupyter Notebooks are also available

From the book “Cloud Computing for Science and Engineering” by Ian Foster and Dennis Gannon, to be published soon by MIT Press.

The link to the book is here

<https://Cloud4SciEng.org>

# Python and Jupyter

Jupyter in the cloud

Go to <https://notebooks.azure.com>

Signup – it's free.

If you are new to Jupyter do

Welcome.ipynb

If you are new to Python do Python.ipynb

Installing Jupyter on your laptop

Go to <https://docs.continuum.io/anaconda/install>

Do it. Then “Jupyter notebook” at the shell

A Better solution: install Docker

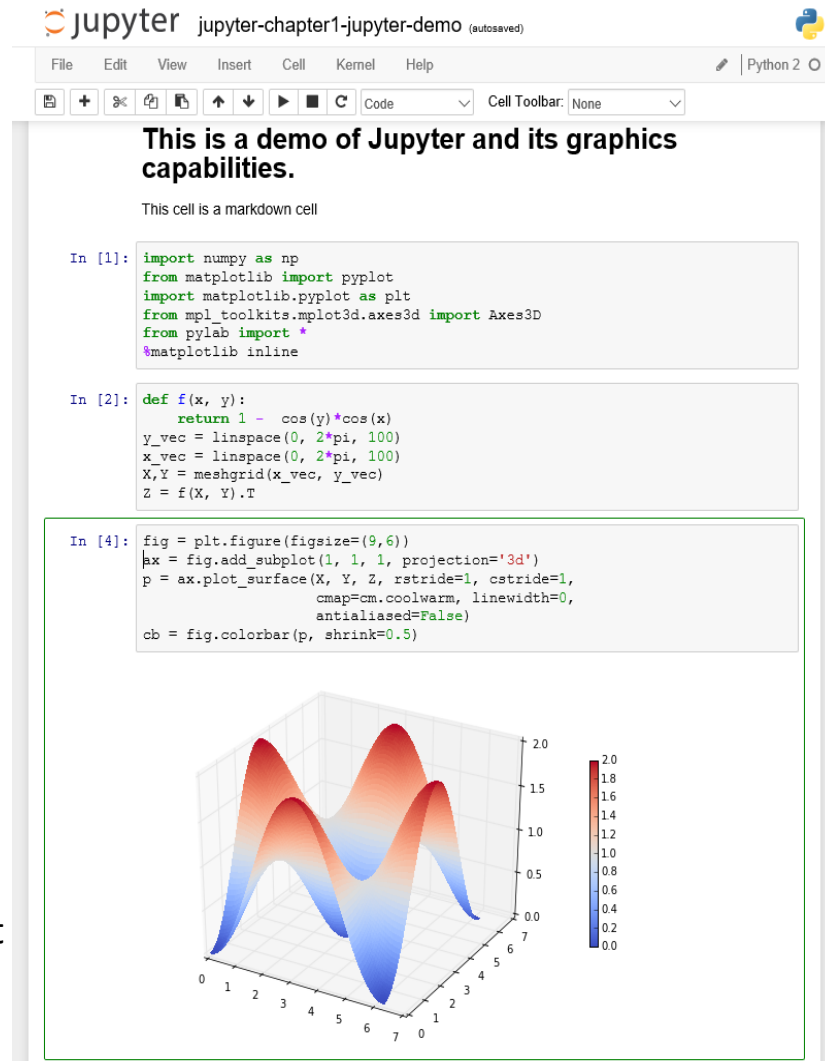
<https://docs.docker.com/engine/installation/>

and run Jupyter in a container

`docker run -it -p 8888:8888 dbgannon/tutorial`

Accept security exceptions and login with “tutorial”

Open notebook jupyter.ipynb to see the one on the right





# Two ways to access the cloud

## Portals and SDKs

### Web Portals

Dashboard that allow you to see and manage your cloud resources.

### Software Development Kits (SDKs)

Libraries that give you the tools to manage cloud resources from a program or script.

Based on REST web service calls

Let's look at several Cloud Web Portals

# Amazon AWS Portal

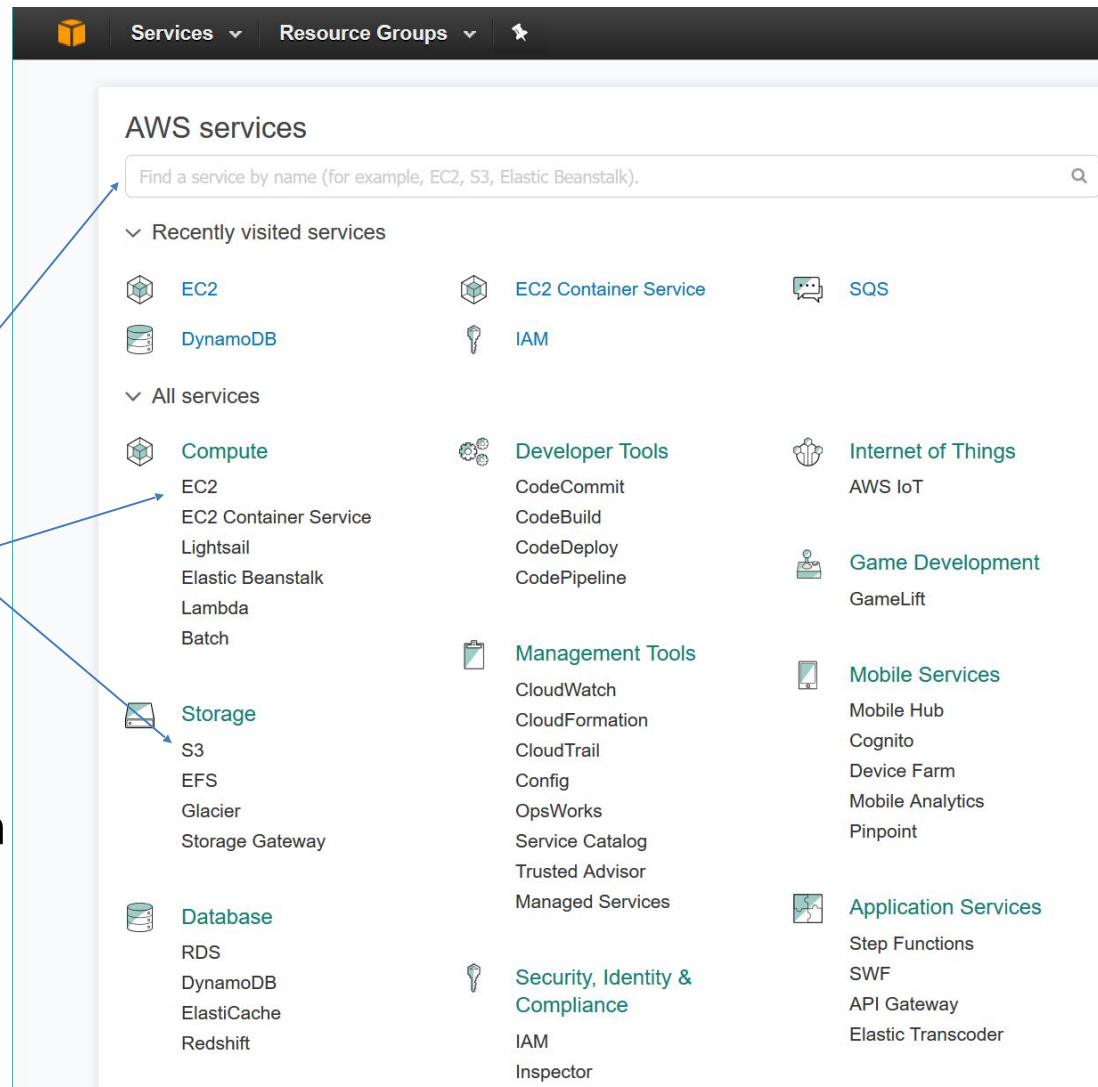
You see links to all the standard services

You also have a search bar to find others.

To create a storage account go to S3

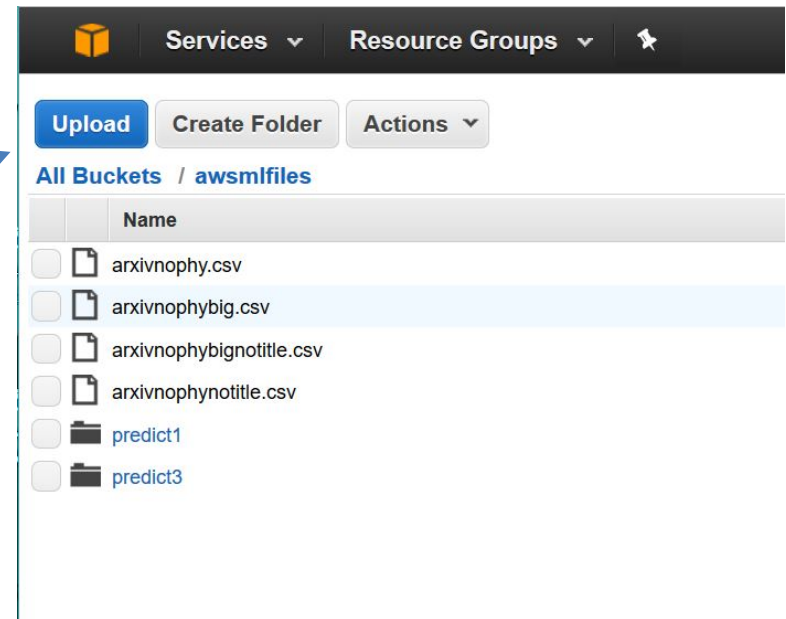
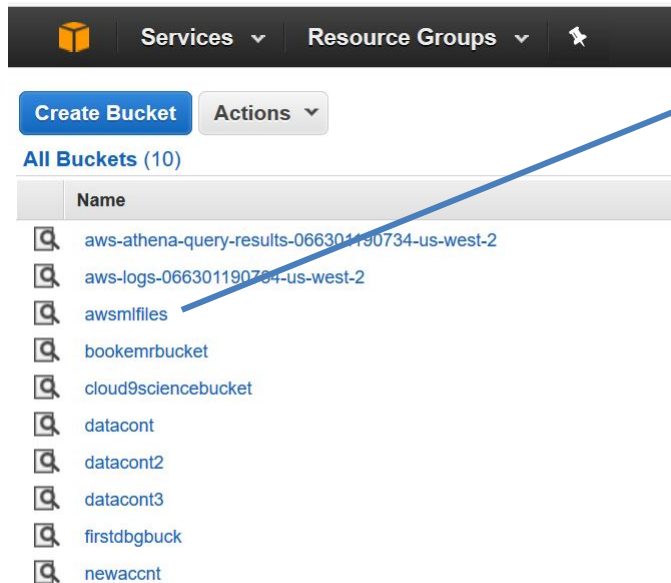
To launch a Virtual Machine go to EC2

Let's look at the S3 storage system



# Selecting S3 we get the bucket list

Selection “awsmifiles” gives us the a list of the contents



Notice that it has regular objects AND folders.

# AWS: Enticing New Customers

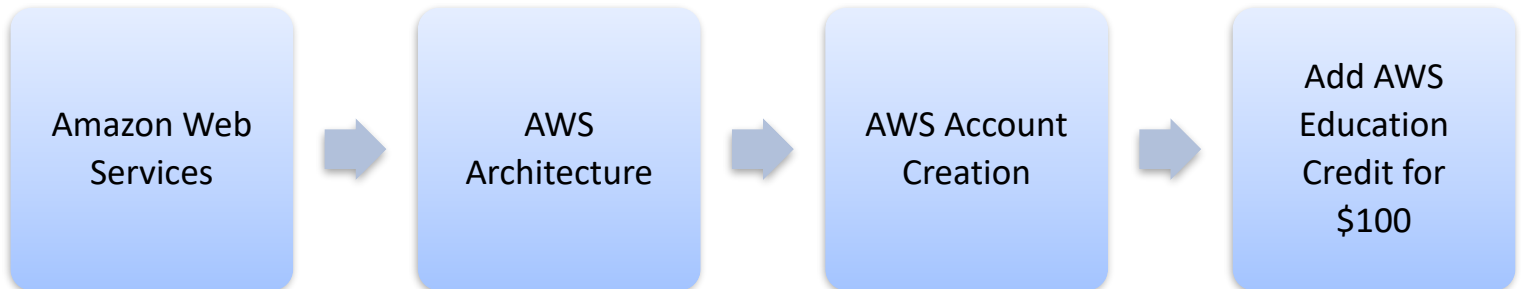
## Free Tier\*

As part of [AWS's Free Usage Tier](#), new AWS customers can get started with Amazon EC2 for free. Upon sign-up, new AWS customers receive the following EC2 services each month for one year:

- 750 hours of EC2 running Linux/Unix Micro instance usage
- 750 hours of EC2 running Microsoft Windows Server Micro instance usage
- 750 hours of Elastic Load Balancing plus 15 GB data processing
- 30 GB of Amazon Elastic Block Storage (EBS) plus 2 million IOs and 1 GB snapshot storage
- 15 GB of bandwidth out aggregated across all AWS services
- 1 GB of Regional Data Transfer



# Lab Steps



## **Lab – AWS Account Setup, Services Overview, Resource Discovery, and Instance Setup**

### **Purpose of the Lab**

Understand definitions of various Amazon Web Services (AWS) and their use in cloud computing based web applications that are accessible over the Internet through an AWS account

Use the AWS account for the discovery, reservation and access of virtual compute/storage infrastructure instances



# Watch out for unexpected Costs

When you finish your work remember to make sure of the following to **avoid unwanted costs**:

- Delete** your **S3** objects.

- Stop or **Shut Down** your **EC2** and **RDS** instances.

The customer is responsible for the resources she's using.

AWS **declines any responsibility** if the customer forgets to shut down resources.

# The Azure Portal

Select "+ New"

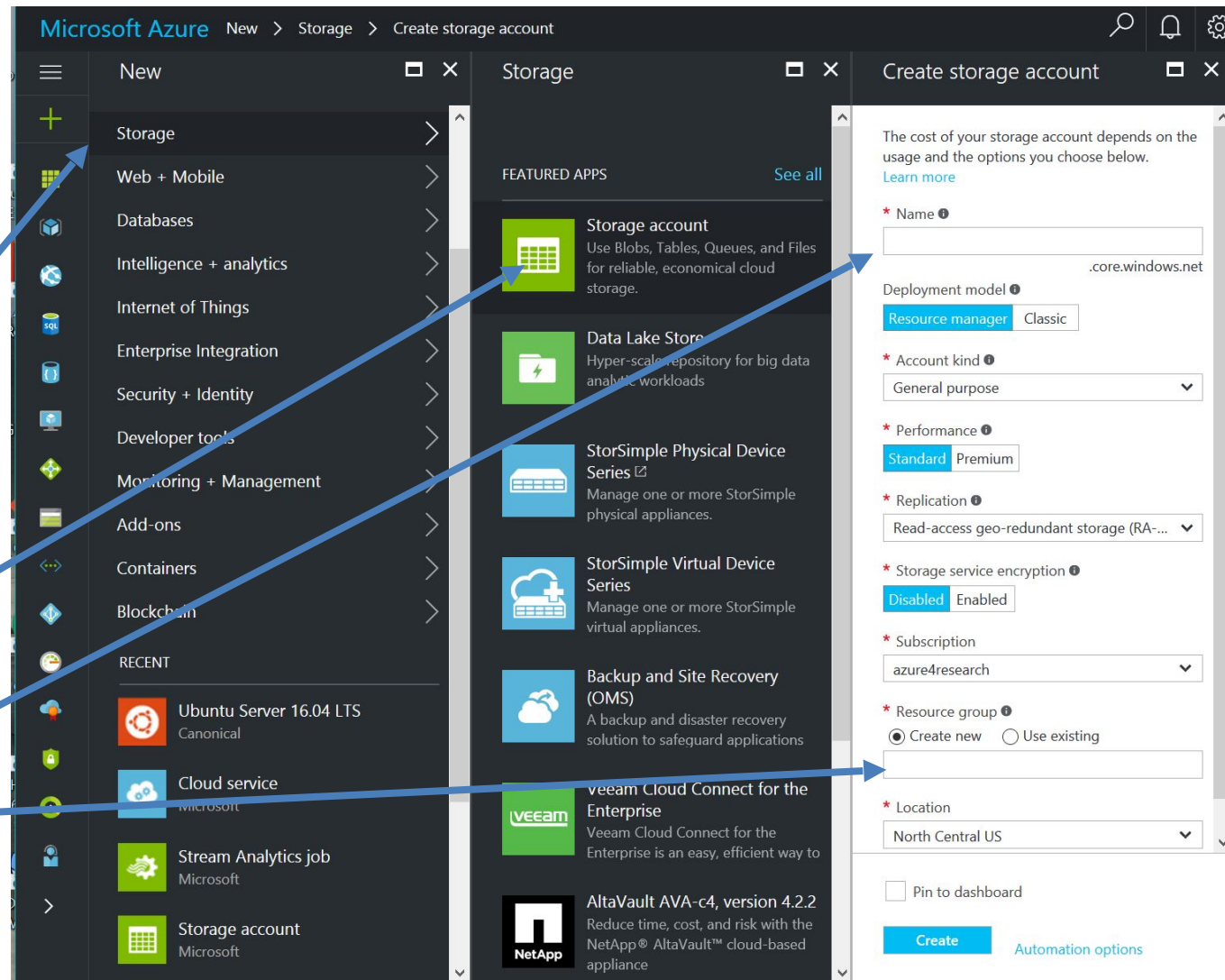
The screenshot shows the Microsoft Azure Portal interface. The left sidebar contains a navigation menu with the following items: **New** (highlighted with a blue arrow), All resources, Resource groups, App Services, SQL databases, NoSQL (DocumentDB), Virtual machines, Load balancers, Storage accounts, Virtual networks, Azure Active Directory, Monitor, Azure Advisor, Security Center, Billing, and Help + support. The main content area is titled 'Dashboard' and includes a search bar and several action buttons (New dashboard, Edit dashboard, Share, Fullscreen, Clone, Delete). The dashboard is divided into three main sections:   
1. **All resources** (ALL SUBSCRIPTIONS): A table listing resources such as biggerMLVM (Virtual machine), biggerMLVM-nsg (Network security group), biggermlvm842 (Network interface), scimldata (Storage account), predicttopicPlan (Machine Learning Web Ser...), predicttopicstorage (Storage account), escistore2 (Storage account), bookcluster3store (Storage account), and bookcluster2store (Storage account).   
2. **Get started**: A section with cards for Virtual Machines, App Service, SQL Database, Storage, and Azure Portal.   
3. **Resources BOOKRG**: A section showing a list of resources including biggerML, biggerMLVM, and biggerMLVM.   
At the bottom of the dashboard, there is a **Service health** section (MY RESOURCES) displaying a world map with status indicators, and a **Marketplace** section.

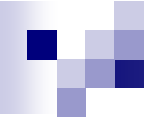
Selecting “+” gives this list of options.

Selecting “Storage” gives the secondary menu of types of storage apps.

To create a storage account select the top one.

- give it a name, research group and location.





# Web Sources

- [cs.queensu.ca](#) › [~khalifa](#) › [AWS1](#)
- <https://media.amazonwebservices.com> › [icons](#) › [AWS\\_Simple\\_Icons\\_ppt](#)
- [cs.uccs.edu](#) › [aws](#)
- [silver.web.unc.edu](#) › [files](#) › [2015/07](#) › [AWS-Slides](#)
- <https://cse.buffalo.edu> › [~bina](#) › [cse487](#) › [fall2010](#) › [AWSOct18](#)
- <https://d36cz9buwru1tt.cloudfront.net> › [AWS\\_Overview](#)



# Other Interesting Links

- <http://docs.amazonwebservices.com/AWSEC2/latest/UserGuide/>
- <http://docs.amazonwebservices.com/AWSEC2/latest/CommandLineReference/>
- <http://docs.amazonwebservices.com/AWSEC2/latest/APIReference/>

